

1 Package guess

1.1 Introduction to package guess

Package **guess** guesses a formula for a sequence of numbers.

This package provides functions to find a formula for a sequence, of numbers within the hierarchy of expressions of the form, <rational function>, <product of rational function>, <product of product of rational function>, etc.

This package is a translation of the Mathematica package **Rate.m** by Christian Krattenthaler (Kratt at pap.univie.ac.at). The translation to Maple was done by Jean-Francois Beraud (Jean-Francois.Beraud at sic.sp2mi.univ-poitiers.fr) and Bruno Gauthier (Bruno.Gauthier at univ-mlv.fr).

All features of this package are due to C. Krattenthaler. The help text is due to Jean-Francois Beraud and Bruno Gauthier.

1.2 Definitions for package guess

guess [Function]

```
guess(1)
guess(1, nlevels)
guess(1, "one")
guess(1, "nogamma")
```

guess(1) tries to find a formula for a sequence within the hierarchy, of expressions of the form <rational function>, <product of rational function>, <product of product of rational function>, etc.

guess(1, nlevels), where *nlevels* is a positive integer, does the same thing as **guess(1)**, but it searches only within the first *nlevels* levels.

guess(1, "one") does the same thing as **guess(1)** but it returns the first solution it finds.

guess(1, "nogamma") does the same thing as **guess(1)** but it returns expressions without gamma functions.

Examples:

```
guess([1,2,3]);
```

[i0]

```
guess([1,4,9,16]);
```

$$\frac{1}{2}$$

[i0]

```
guess([1,2,6,24,120]);
```

$$\frac{i0 - 1}{==}\backslash$$

! !

```

[ ! ! (i1 + 1)]
! !
i1 = 1
guess(makelist(product(product(gamma(i)*i^2,i,1,j),j,1,k),k,1,8));

```

```

i0 - 1 i1 - 1 i2 - 1
/===\ /===\ /===\ 2
! ! ! ! (i3 + 3)
[ ! ! 4 ! ! 18 ! ! -----]
! ! ! ! i3 + 2
i1 = 1 i2 = 1 i3 = 1
guess([1,2,7,42,429,7436,218348,10850216]);

```

```

i0 - 1 i1 - 1
/===\ /===\
! ! ! ! 3 (3 i2 + 2) (3 i2 + 4)
[ ! ! 2 ! ! -----]
! ! ! ! 4 (2 i2 + 1) (2 i2 + 3)
i1 = 1 i2 = 1
guess(makelist(k^3+k^2,k,1,7));

```

Dependent equations eliminated: (6)

```

i0 - 1
/===\
2
[i0 (i0 + 1), 2 ! ! (- -----),
! ! 5040
! ! 4 3 2
i1 = 1 i1 - 24 i1 + 245 i1 - 1422 i1 + 360]

```

```

i0 - 1
/===\
! ! (i1 + 1) (i1 + 2)
2 ! ! -----]
! ! 2
i1 = 1 i1

```

Note that the last example produces three solutions. The first and the last are equivalent, but the second is different! In this case,

```
guess(makelist(k^3+k^2,k,1,7),1);
```

or

```
guess(makelist(k^3+k^2,k,1,7),"one");
```

find only the solution $i0^2*(i0 + 1)$, which is a rational function, and is also the first function `guess` finds.

Appendix A Function and variable index

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(Index is nonexistent)